

SODS Final Project - Group 21

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1 Baseline Synthesis

Table 1: Baseline synthesis results

Metric	Value
Clock period (ns)	2.0
Area (μm^2)	38381.33
Slack (ns)	0.000203
Leakage power (mW)	1.365×10^{-5}
Dynamic power (mW)	5.930×10^{-3}

2 Power Optimization

2.1 Clock-Gating

Table 2: Clock-gating optimization results

Metric	Value
Area (μm^2)	28276.83
Slack (ns)	0.000000
Leakage power (mW)	8.637×10^{-6}
Dynamic power (mW)	4.437×10^{-3}

- **Assumptions:** Only Low-VT cells are used, based on the project requirement. Using `compile_ultra` instead of `compile` according to the labs.
- **Dynamic Power (decreased significantly):** Dynamic power decreases from 5.930 mW in the baseline synthesis to 4.437 mW after clock-gating optimization, corresponding to a reduction of about 25.2%. Clock gating reduces unnecessary switching activity by preventing registers or register groups from toggling when their enable condition is inactive. However, the PrimeTime power breakdown shows that the largest measured reduction comes from the combinational logic. The combinational total power decreases from 3.238 mW to 1.819 mW, while the clock-network power decreases from 1.420 mW to 1.279 mW. This is consistent with the DC area report, which shows a large reduction in combinational cells, nets, and buffer/inverter cells. Therefore, the dynamic-power reduction is due to both reduced switching activity from clock gating and the smaller optimized implementation produced by `compile_ultra`.
- **Area (decreased drastically):** The total cell area decreases from $38382.24 \mu m^2$ to $28276.56 \mu m^2$, corresponding to a reduction of about 26.3%. The DC area report shows that the number of cells decreases from 10580 to 5945, and the number of combinational cells decreases from 9377 to 4846. The number of buffer/inverter cells also decreases strongly, from 3409 to 652. This indicates that the clock-gated implementation is structurally much smaller than the baseline. The reduction can be explained by the combined

effect of clock-gating insertion, possible replacement of register-enable feedback multiplexers with shared integrated clock-gating cells, and the more aggressive restructuring and optimization performed by `compile_ultra`.

- **Static Power (decreased significantly):** Leakage power decreases from 1.365×10^{-2} mW to 8.637×10^{-3} mW, corresponding to a reduction of about 36.7%. Since both implementations use only LVT cells, this reduction is not due to threshold-voltage optimization. Instead, it is mainly due to the smaller implementation after clock-gating and `compile_ultra` optimization. The reduced number of cells and smaller total cell area lead to fewer leaking devices, reducing the total leakage power.
- **Slack (remained approximately zero):** The difference between the slack in the baseline synthesis and in the clock-gated implementation is very small, and both designs can be considered to meet timing with almost zero slack. This result is expected because `compile_ultra`, used in this section, performs more aggressive area and power optimization than the `compile` command used in the baseline synthesis, while still satisfying the timing constraint. Therefore, the tool may reduce unnecessary positive slack by resizing or restructuring non-critical logic, leaving the design close to the timing boundary.

2.2 Clock-Gating + Multi-VT

Table 3: Clock-gating and Multi-VT optimization results

Metric	Value
Percentage of LVT cells (%)	10.53
Percentage of SVT cells (%)	7.15
Percentage of HVT cells (%)	82.32
Area (μm^2)	31294.98
Slack (ns)	0.000000
Leakage power (mW)	2.180×10^{-6}
Dynamic power (mW)	4.852×10^{-3}

- **Area increased:** The total cell area increases about (10.7%). The DC area report shows that the number of cells increases from 5945 to 6858, while the number of sequential cells remains unchanged at 1066. Therefore, the additional cells are entirely combinational. In particular, the number of combinational cells increases from 4846 to 5759, and the number of buffer/inverter cells increases from 652 to 859. This indicates that the synthesis tool added or resized combinational logic, including extra buffers and inverters, to recover timing after replacing many fast LVT cells with slower SVT/HVT cells.

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- **Dynamic power increased:** The dynamic power increases corresponding to an increase of about 9.35%. The PrimeTime power breakdown shows that this increase is mainly due to the combinational logic. The combinational total power increases from 1.819 mW to 2.310 mW, while the clock-network and register power both slightly decrease. More specifically, the combinational switching power increases from 1.119 mW to 1.496 mW. Therefore, the increase in dynamic power is mainly caused by the larger switched capacitance of the combinational logic. This is consistent with the higher number of cells, nets, buffers, and inverters in the Multi-VT implementation and the formula : $P_{dyn} = \alpha \cdot C_L \cdot V_{DD}^2 \cdot f$.
- **Static power decreased:** The leakage power decreases about 74.8%. This is the expected result of Multi-VT optimization. In the clock-gating-only implementation, the design uses only LVT cells, which are fast but have higher leakage current. In the Multi-VT implementation, most of the design is mapped to HVT cells, with only a small percentage

of LVT cells kept on timing-critical paths. Since HVT cells have a higher threshold voltage, their subthreshold leakage current is much lower. Therefore, replacing most LVT cells with HVT/SVT cells significantly reduces static power.

- **Slack stayed approximately zero:** In both implementations, the slack is approximately 0ns, meaning that the timing constraint is just met. This is expected because `compile_ultra` optimizes the design while satisfying the clock-period constraint. In the Multi-VT case, the tool uses slower HVT/SVT cells where possible to reduce leakage, while compensating on timing-critical paths by using additional or stronger combinational logic. As a result, leakage power is reduced, but the design remains close to the timing boundary.

Appendix : Supporting Reports for Clock-Gating and Multi-VT Discussions

Table 4: Evidence from DC and PrimeTime reports comparing baseline synthesis and clock-gating optimization

Source	Metric	Baseline	Ex2.1	Change	Interpretation
DC area report	Number of nets	11097	6771	-4326	Fewer nets reduce switched capacitance.
DC area report	Number of cells	10580	5945	-4635	Clock-gated implementation is much smaller.
DC area report	Combinational cells	9377	4846	-4531	Large reduction in combinational logic.
DC area report	Sequential cells	1203	1066	-137	Sequential cell count also decreases.
DC area report	Buffer/inverter cells	3409	652	-2757	Large reduction in buffer/inverter logic.
DC area report	Total cell area (μm^2)	38382.24	28276.56	-10105.68	Area decreases by about 26.3%.
DC area report	Combinational area (μm^2)	25634.44	16904.16	-8730.28	Area reduction is mainly combinational.
DC area report	Buffer/inverter area (μm^2)	5491.20	1079.00	-4412.20	Buffer/inverter area decreases significantly.
PT power report	Internal power (μW)	3.369	2.326	-1.043	Cell internal dynamic power decreases.
PT power report	Net switching power (μW)	2.560	2.111	-0.449	Net switching power decreases.
PT power report	Dynamic power (μW)	5.929	4.437	-1.492	Dynamic power decreases by about 25.2%.
PT power report	Leakage power (μW)	0.01365	0.008637	-0.005013	Leakage decreases by about 36.7%.
PT power report	Total power (μW)	5.943	4.445	-1.498	Total power decreases mainly due to dynamic power.
PT power groups	Clock-network total power (μW)	1.420	1.279	-0.141	Clock-network power decreases slightly.
PT power groups	Register total power (μW)	1.285	1.347	+0.062	Register power slightly increases.
PT power groups	Combinational total power (μW)	3.238	1.819	-1.419	Largest power reduction comes from combinational logic.
PT power groups	Combinational switching power (μW)	2.126	1.119	-1.007	Strong evidence of reduced switched capacitance.

Table 5: Evidence from DC and PrimeTime reports comparing clock-gating-only and clock-gating + Multi-VT implementations

Source	Metric	Ex2.1	Ex2.2	Change	Interpretation
DC area report	Number of nets	6771	7816	+1045	More nets increase switched capacitance.
DC area report	Number of cells	5945	6858	+913	Multi-VT implementation uses more cells.
DC area report	Combinational cells	4846	5759	+913	Increase is entirely in combinational logic.
DC area report	Sequential cells	1066	1066	0	Register count is unchanged.
DC area report	Buffer/inverter cells	652	859	+207	Extra buffering contributes to capacitance.
DC area report	Total cell area (μm^2)	28276.56	31294.64	+3018.08	Area increases by about 10.7%.
DC area report	Combinational area (μm^2)	16904.16	19754.80	+2850.64	Area increase is mainly combinational.
DC area report	Buffer/inverter area (μm^2)	1079.00	1445.60	+366.60	Buffer/inverter area increases significantly.
PT power report	Internal power (μW)	2.326	2.413	+0.087	Cell internal dynamic power increases slightly.
PT power report	Net switching power (μW)	2.111	2.439	+0.328	Main contributor to dynamic power increase.
PT power report	Leakage power (μW)	0.008637	0.002180	-0.006457	Leakage decreases by about 74.8%.
PT power report	Total power (μW)	4.445	4.854	+0.409	Total power increases due to dynamic power.
PT power groups	Clock-network total power (μW)	1.279	1.235	-0.044	Clock network power decreases slightly.
PT power groups	Register total power (μW)	1.347	1.309	-0.038	Register power decreases slightly.
PT power groups	Combinational total power (μW)	1.819	2.310	+0.491	Dynamic increase comes from combinational logic.
PT power groups	Combinational switching power (μW)	1.119	1.496	+0.377	Strong evidence of larger switched capacitance.
DC reference report	Full-adder cells	240	371	+131	Mapped combinational logic becomes larger.
DC reference report	NAND cells	850	1102	+252	More combinational gates are used.
DC reference report	NOR cells	336	524	+188	More combinational gates are used.
DC reference report	Inverter cells	594	780	+186	More inverters are inserted.
DC reference report	Buffer cells	58	79	+21	More buffers are inserted.
DC reference report	Sequential cells	1033	1033	0	Sequential mapping is unchanged.